Appendix A: Tag-length-value (TLV) Encoding Format

A.1. Scope & Purpose

The Matter TLV (*Tag-Length-Value*) format is a generalized encoding method for simple structured data, used throughout this specification.

Values in the Matter TLV format are encoded as *TLV elements*. Each TLV element has a type. Element types fall into two categories: *primitive types* and *container types*. Primitive types convey fundamental data values such as integers and strings. Container types convey collections of elements that themselves are either primitives or containers. The Matter TLV format supports three different container types: structures, arrays and lists.

All valid *TLV encodings* consist of a single top-level element. This value can be either a primitive type or a container type.

A.2. Tags

A TLV element includes an optional numeric tag that identifies its purpose. A TLV element without a tag is called an *anonymous* element. For elements with tags, two categories of tags are defined: *profile-specific* and *context-specific*.

A.2.1. Profile-Specific Tags

Profile-specific tags identify elements globally. A profile-specific tag is a 64-bit number composed of the following fields:

- 16-bit Vendor ID
- 16-bit profile number
- 32-bit tag number

Profile-specific tags are defined either by Matter or by vendors. Additionally the Matter Common Profile includes a set of predefined profile-specific tags that can be used across organizations.

A.2.2. Context-Specific Tags

Context-specific tags identify elements within the context of a containing structure element. A context-specific tag consists of a single 8-bit tag number. The meaning of a context-specific tag derives from the structure it resides in, implying that the same tag number may have different meanings in the context of different structures. Effectively, the interpretation of a context-specific tag depends on the tag attached to the containing element. Because structures themselves can be assigned context-specific tags, the interpretation of a context-specific tag may ultimately depend on a nested chain of such tags.

Context-specific tags can only be assigned to elements that are immediately within a structure. This

